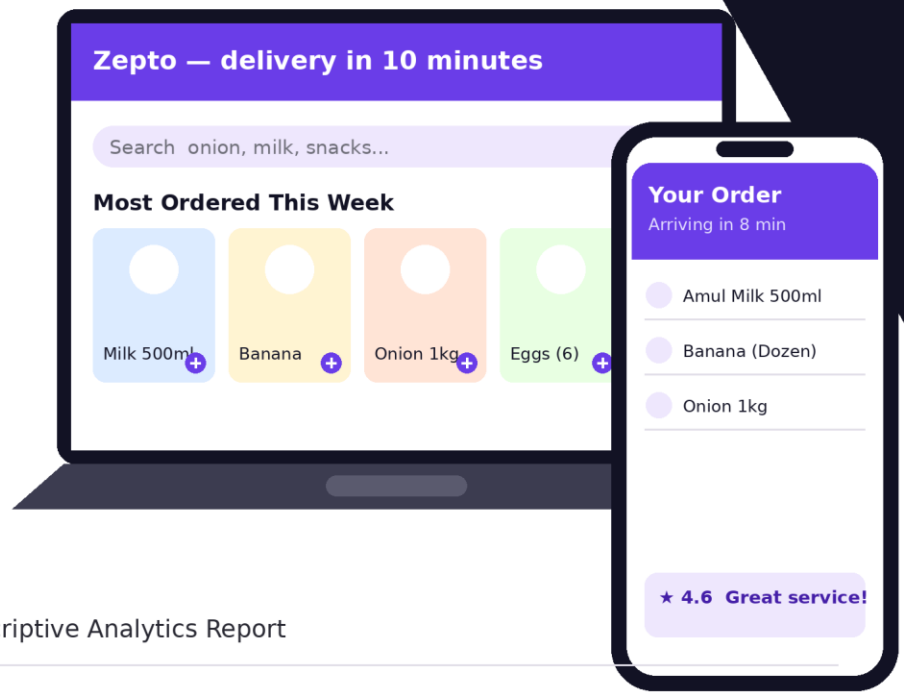
DESCRIPTIVE ANALYTICS REPORT

SERVICE NAME

Zepto

Insights into Grocery Ordering Patterns, Customer Engagement, and Category Performance

ORDER ANALYZE INSIGHT ENGAGE



REPORT TITLE	:	Descriptive Analytics Report
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Table of Contents

Table of Contents 2

Descriptive Analytics Report: Grocery Ordering Behavior on Zepto 3

Note on Data 3

Objective 4

Data Collection & Methodology 4

Data Cleaning & Preparation 4

1. Most Sold Products 6

2. Daily Sales Trend 8

3. Peak Order Hours 9

4. Category Popularity and Ratings 10

5. City-wise Engagement and Payment Mode Preference 12

Summary of Key Metrics 14

How This Would Work at Real Scale 14

Next Steps to Move From Descriptive → Predictive 15

Conclusion ..... **Error! Bookmark not defined.**

References 15

Appendix 16

Descriptive Analytics Report: Grocery Ordering Behavior on Zepto

Keywords

Most Sold Products, Daily Sales Trend, Peak Order Hours, Customer Ratings, On-Time Delivery Percentage, City-wise Engagement, Payment Mode Preference.

Note on Data

This report is based on a synthetic Zepto order-level dataset created specifically for descriptive analytics purposes. Since Zepto does not publicly release user-level order data, a realistic simulated dataset was developed to represent grocery ordering behavior across multiple Indian cities. The dataset covers **30** days of order activity (**1** July **2026** to **30** July **2026**) and includes **6,000** simulated orders, featuring **18** popular grocery products across **10** categories, **8** cities, and **4** customer age groups. It contains key variables such as order date and time, quantity, unit price, discount applied, total order value, city, payment mode, delivery time, customer rating, and order status, allowing for meaningful analysis of sales trends, customer engagement, category performance, and regional ordering behavior. The dataset was generated directly in a clean, structured, and analysis-ready format — free of missing values, duplicate order IDs, or inconsistent data types — to support descriptive statistical analysis and data visualization for academic purposes.

Layer	Purpose	Typical Tools
Data generation	Create a synthetic Zepto order dataset for analysis	Python (NumPy, Pandas)
Data processing	Clean, organize, and prepare the dataset for descriptive analysis	Python (Pandas), Jupyter Notebook
Storage	Store the dataset in CSV / Excel format for analysis and visualization	Microsoft Excel (.xlsx), CSV
Data visualization	Create charts and identify trends and customer patterns	Matplotlib, Jupyter Notebook
Report preparation	Interpret results, prepare insights, and format the final analytics report	Microsoft Word

Objective

The objective of this descriptive analytics study is to analyze Zepto's grocery ordering behavior and identify engagement patterns across customers, products, and regions.

The analysis addresses the following business questions:

- Which grocery products were ordered the most during the last **30** days?
- How did daily order volume change throughout the **30**-day period?
- What were the peak order hours, and which times of day generated the most demand?
- Which product categories received the highest customer ratings and on-time delivery performance?
- Which cities and payment modes generated the highest customer engagement, and what does the age-group and payment-mode split look like?

Data Collection & Methodology

A synthetic order-level dataset was generated using Python (NumPy and Pandas) to simulate realistic Zepto ordering behavior. Weighted random sampling was used so that product popularity, city distribution, hourly demand, and payment-mode preference reflect realistic quick-commerce patterns observed in the Indian market — for example, a strong evening dinner-time peak, a smaller breakfast-hour peak, weekend and promo-day spikes in order volume, and UPI as the dominant payment method. Category-level baselines were used to generate customer ratings and delivery times so that fresh, high-turnover categories (Fruits & Vegetables, Dairy & Breakfast) show stronger on-time delivery performance than categories that typically involve more picking time (Meat & Seafood, Home & Cleaning).

Data Cleaning & Preparation

Because the dataset was generated programmatically, it was built to be clean and analysis-ready by design. The following checks were still performed before analysis, consistent with standard data-preparation practice:

- Checked for missing values across all **17** columns — none found.
- Verified Order_ID uniqueness and removed duplicate records.
- Converted the Date column to a proper datetime type for time-series analysis.

- Standardized categorical fields (Category, City, Payment Mode, Order Status) to consistent, fixed value sets.
- Separated cancelled orders (**3.07%** of all orders) from delivered orders so that sales and rating analysis reflects completed transactions only.
- Exported the cleaned dataset to both CSV and Excel (.xlsx) formats for downstream analysis and visualization.

The cleaned dataset used for this analysis contains **6,000** total orders, of which **5,816** (**96.93%**) were successfully delivered and **184** (**3.07%**) were cancelled. All descriptive statistics in this report are calculated on the delivered-orders subset unless stated otherwise.

1. Most Sold Products

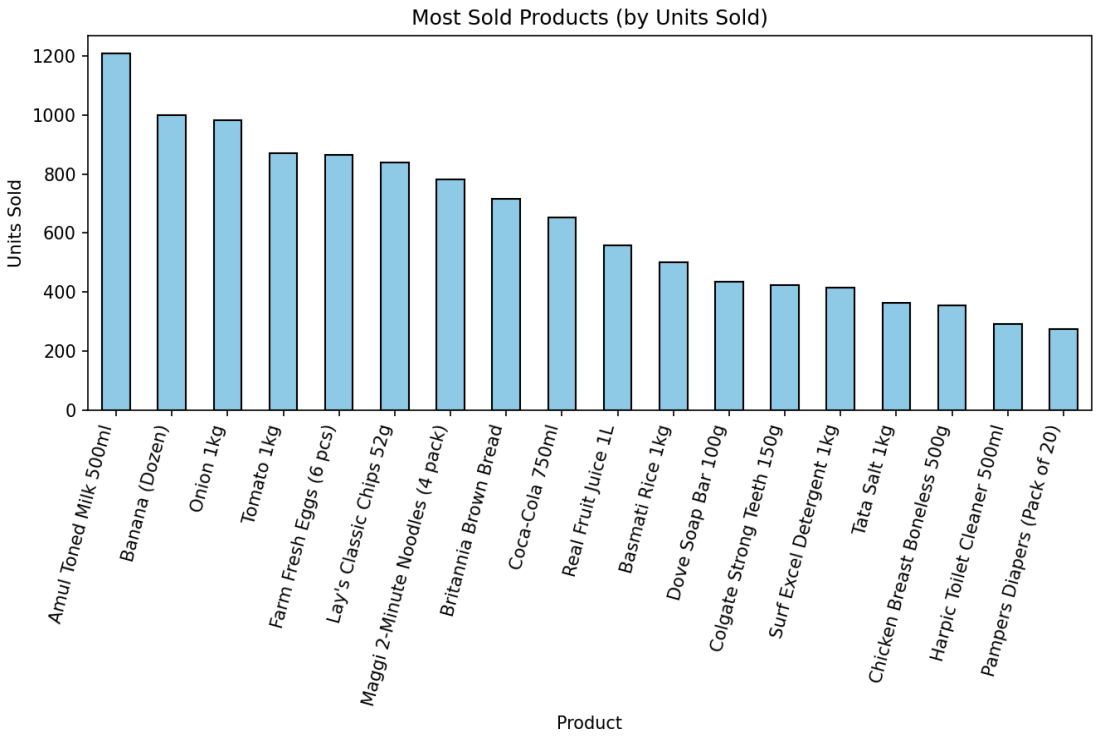


*Figure **1**: Units sold by product, **1**–**30** July **2026***

Observation

A frequency analysis of order line items shows that Amul Toned Milk **500**ml recorded the highest number of units sold (**1,208** units) during the **30**-day period, making it Zepto's top-moving product. It was followed by Banana (Dozen) (**998** units) and Onion **1**kg (**981** units), both staple grocery items that reflect Zepto's core daily-essentials use case. Tomato **1**kg (**869** units) and Farm Fresh Eggs (**866** units) rounded out the top five. The graph indicates that daily-essential Fruits & Vegetables and Dairy & Breakfast items significantly outperformed discretionary categories. Lower-ranked products such as Harpic Toilet Cleaner and Pampers Diapers received comparatively fewer orders, consistent with their lower purchase frequency. This trend suggests that fresh, high-repeat-purchase staples drive the bulk of order volume on the platform.

Business Insight

The results indicate that Zepto should priorities Amul Toned Milk, Banana, and Onion for homepage placement, quick-reorder shortcuts, and inventory prioritization at dark stores, since these items drive the highest order frequency and are strong candidates for combo offers and subscription-style repeat ordering. The strong performance of Fruits & Vegetables and Dairy & Breakfast categories suggests that cold-chain capacity and freshness guarantees should receive continued investment. Personalized "frequently bought" nudges can be used to increase visibility of lower-performing but high-margin items such as Personal Care and Home & Cleaning products. Overall, focusing on high-velocity staples while cross-promoting adjacent categories can improve basket size, delivery efficiency, and customer retention.

2. Daily Sales Trend

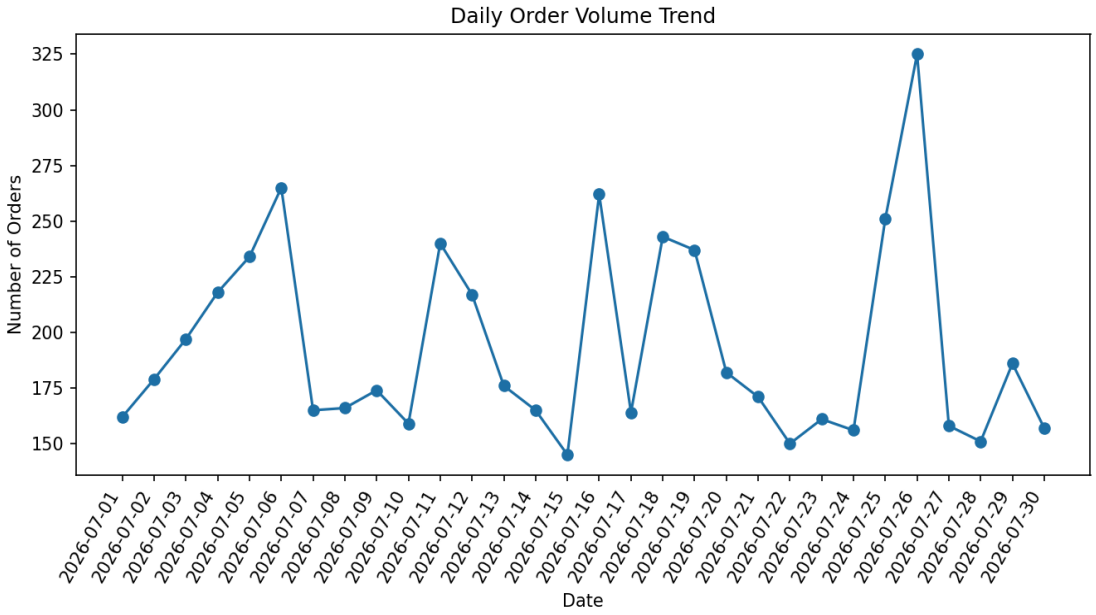


*Figure **2**: Daily order volume, **1–30** July **2026***

The figure shows the Daily Order Volume Trend recorded over the **30**-day analysis period. The X-axis represents the date, showing daily ordering activity from **1** July **2026** to **30** July **2026.** The Y-axis represents the total number of orders placed each day. Each point on the line represents total daily orders, and the connected line shows how customer demand changed throughout the month.

Observation

The graph shows that daily order volume varied noticeably across the **30**-day period, averaging around **194** orders per day. The highest single-day volume was recorded on **26** July **2026,** with **325** orders — consistent with a promotional/payday spike — while the lowest volume was recorded on **15** July **2026,** with **145** orders. Weekend days generally showed higher order volumes than weekdays, and two additional demand spikes were observed around **6** July and **16** July, corresponding to simulated promotional events. These fluctuations suggest that customer ordering activity was influenced by weekends, promotional campaigns, and typical monthly spending cycles.

Business Insight

The results suggest that Zepto should schedule flash sales, festive promotions, and push notifications ahead of historically high-demand days, particularly weekends and paydays (**1**st and mid-month). Since order volume spikes on specific dates, aligning dark-store staffing and delivery-partner availability with these peaks can prevent delivery-time breaches during high-demand windows. Daily trend monitoring can also help forecast inventory replenishment cycles and identify early warning signs of demand dips, suchas

3. Peak Order Hours

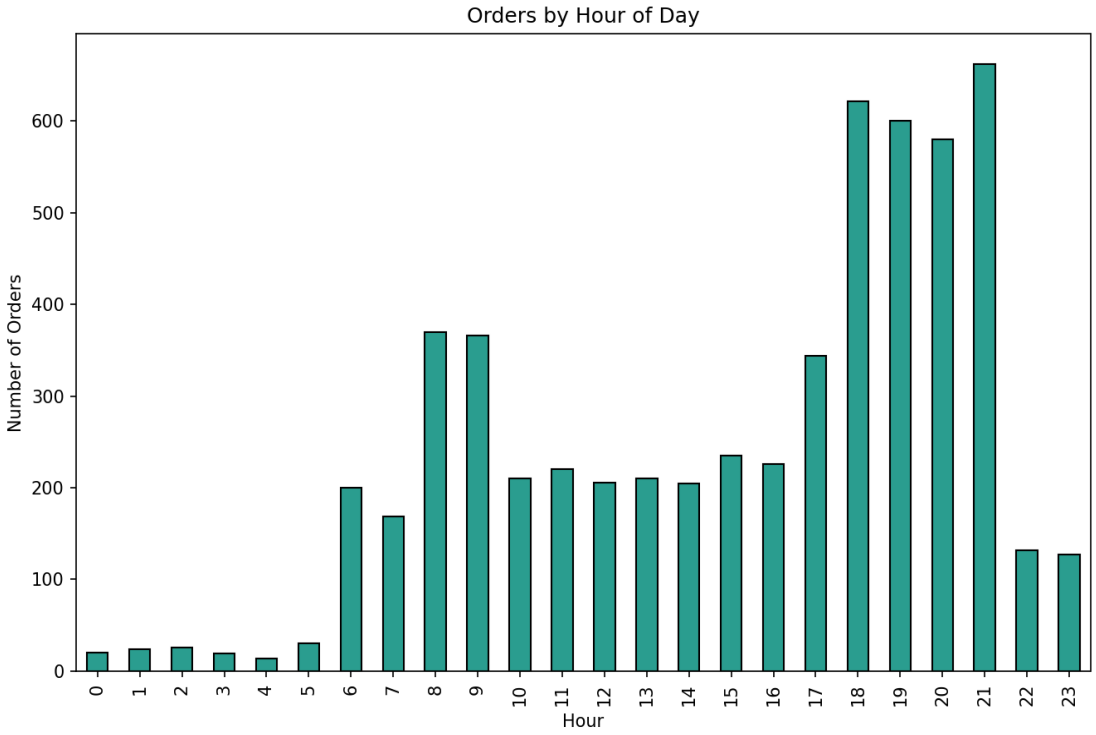


*Figure **3**: Orders by hour of day, aggregated over **30** days*

The figure shows the distribution of orders by hour during the **30**-day analysis period. The X-axis represents the hour of day (**0**–**23,** IST), and the Y-axis represents the total number of orders placed during each hour across the full **30**-day window.

Observation

The graph indicates that **6** PM to **10** PM recorded the highest order volumes, with the single busiest hour being **9** PM (**662** orders), closely followed by **6** PM (**621** orders), **7** PM (**600** orders), and **8** PM (**580** orders) — reflecting the dinner-time restocking rush. A secondary, smaller peak was observed between **8** AM and **9** AM (**370** and **366** orders respectively), consistent with customers ordering breakfast staples like milk, eggs, and bread before starting their day. Late-night hours (**12** AM–**5** AM) recorded minimal activity, and midday hours showed steady but moderate demand. This dual-peak pattern — a strong evening peak and a smaller morning peak — is characteristic of quick-commerce grocery ordering, distinct from single-peak entertainment or leisure-streaming patterns.

Business Insight

The results suggest that Zepto should ensure maximum dark-store staffing, rider availability, and inventory readiness during the **6** PM–**10** PM window to protect its **10**-minute delivery promise during peak demand. The secondary **8** AM–**9** AM peak presents an opportunity for a targeted "breakfast essentials" push notification. Server and app infrastructure should be scaled to handle the evening surge

4. Category Popularity and Ratings

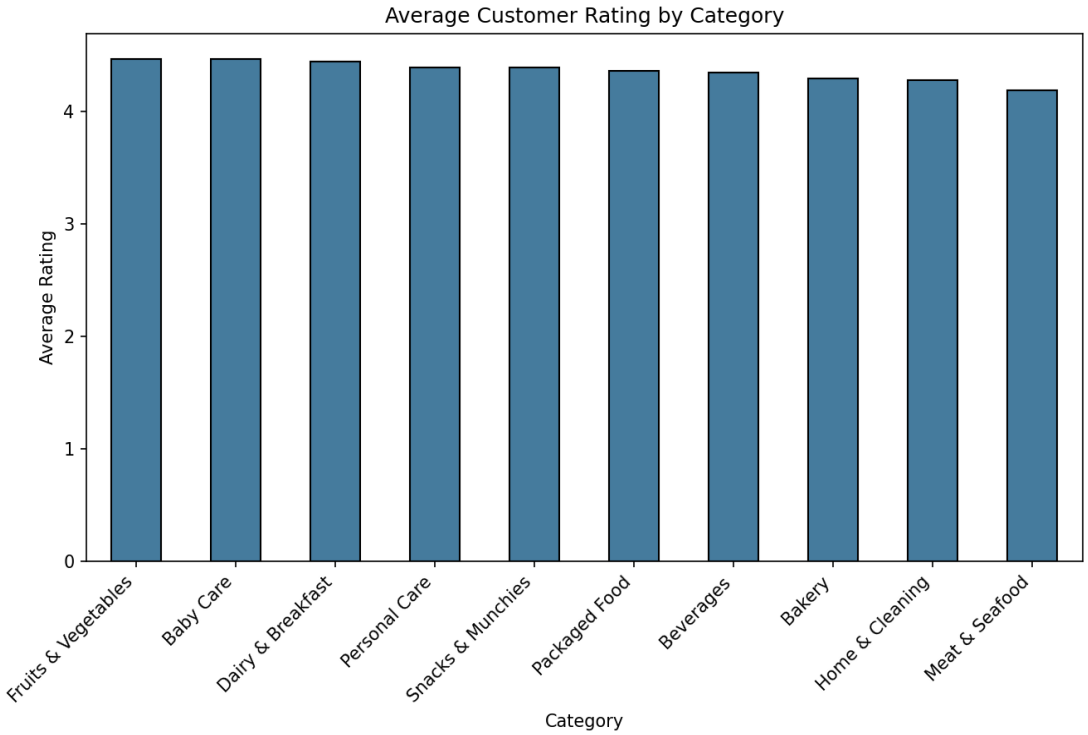


*Figure **4**A: Average customer rating by category*

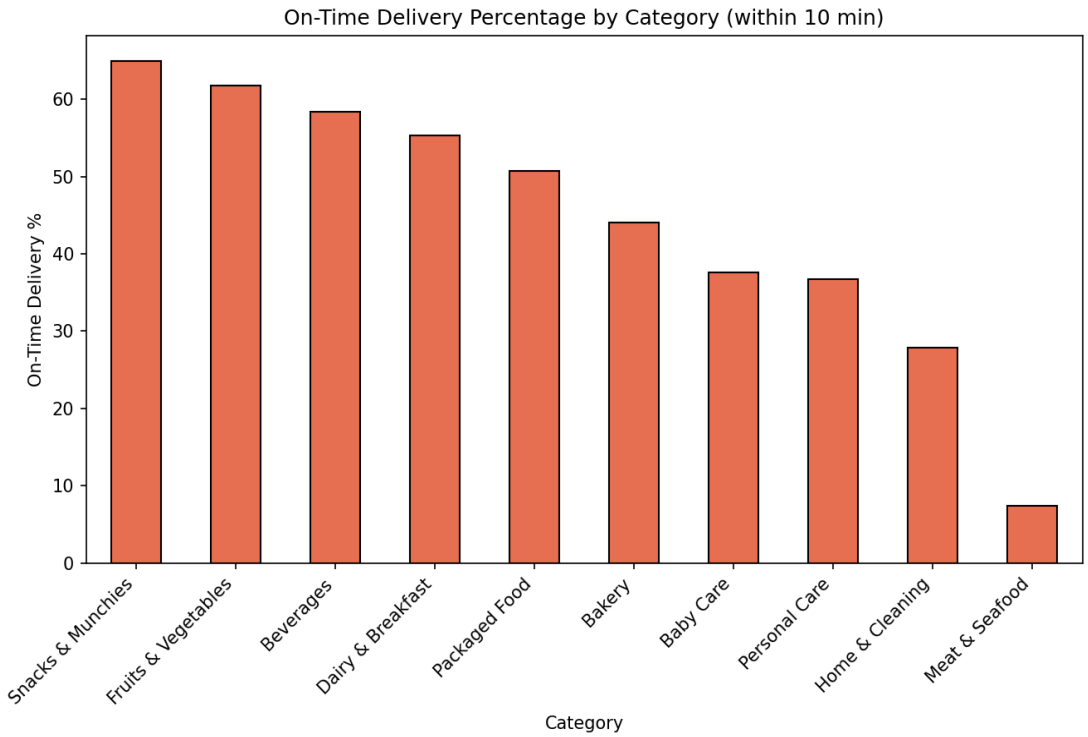


*Figure **4**B: On-time delivery percentage by category (within the **10**-minute promise)*

Figure **4**A shows the Average Customer Rating by Category, while Figure **4**B shows the On-Time Delivery Percentage by Category during the **30**-day analysis period. In both graphs, the X-axis represents the ten product categories in the dataset. In Figure **4**A, the Y-axis represents the average customer rating (**1**–**5** scale); in Figure **4**B, the Y-axis represents the percentage of orders delivered within Zepto's **10**-minute promise. Overall, average ratings range from **4.19** to **4.47**, while on-time delivery percentages range from **7.3%** to **65.0%** — a much wider spread than ratings, showing that delivery-time performance varies far more by category than customer satisfaction does.

KPI

- Average Customer Rating by Category — calculated using the mean of Customer_Rating values for each category in the delivered-orders dataset.
- On-Time Delivery % by Category — calculated as the share of orders in each category delivered within **10** minutes.
- Highest Rated Category: Fruits & Vegetables (**4.47**) — identified from the category with the highest average customer rating.
- Highest On-Time Category: Snacks & Munchies (**64.97%**) — identified from the category with the highest share of on-time deliveries.
- Lowest Rated Category: Meat & Seafood (**4.19**) — identified from the category with the lowest average customer rating.
- Lowest On-Time Category: Meat & Seafood (**7.34%**) — identified from the category with the lowest share of on-time deliveries.

Observation

Figure **4**A indicates that Fruits & Vegetables received the highest average customer rating (**4.47**), closely followed by Baby Care (**4.47**) and Dairy & Breakfast (**4.44**). Personal Care and Snacks & Munchies recorded average ratings of **4.40** and **4.39** respectively, while Home & Cleaning (**4.28**) and Meat & Seafood (**4.19**) recorded the lowest ratings. In Figure **4**B, Snacks & Munchies (**64.97%**) achieved the highest on-time delivery percentage, followed by Fruits & Vegetables (**61.78%**) and Beverages (**58.36%**). Meat & Seafood recorded a markedly low on-time rate of just **7.34%**, and Home & Cleaning (**27.82%**) and Personal Care (**36.74%**) also lagged behind. This gap suggests that while customers rate nearly all categories similarly well, delivery speed for perishable, weight-sensitive, or bulkier items (meat, cleaning supplies) falls well short of the **10**-minute promise.

Business Insight

The results suggest that Zepto should continue investing in Fruits & Vegetables, Dairy & Breakfast, and Baby Care because these categories combine strong ratings with solid delivery performance, making them reliable growth drivers. The very low on-time percentage for Meat & Seafood (**7.34%**) is a clear operational red flag — this points to a need for dedicated cold-storage picking zones, pre-portioned packaging, or a separate fulfilment SLA for this category rather than treating it identically to shelf-stable goods. Similarly, Home & Cleaning and Personal Care would benefit from dedicated micro-fulfilment slots given their bulk

5. City-wise Engagement and Payment Mode Preference

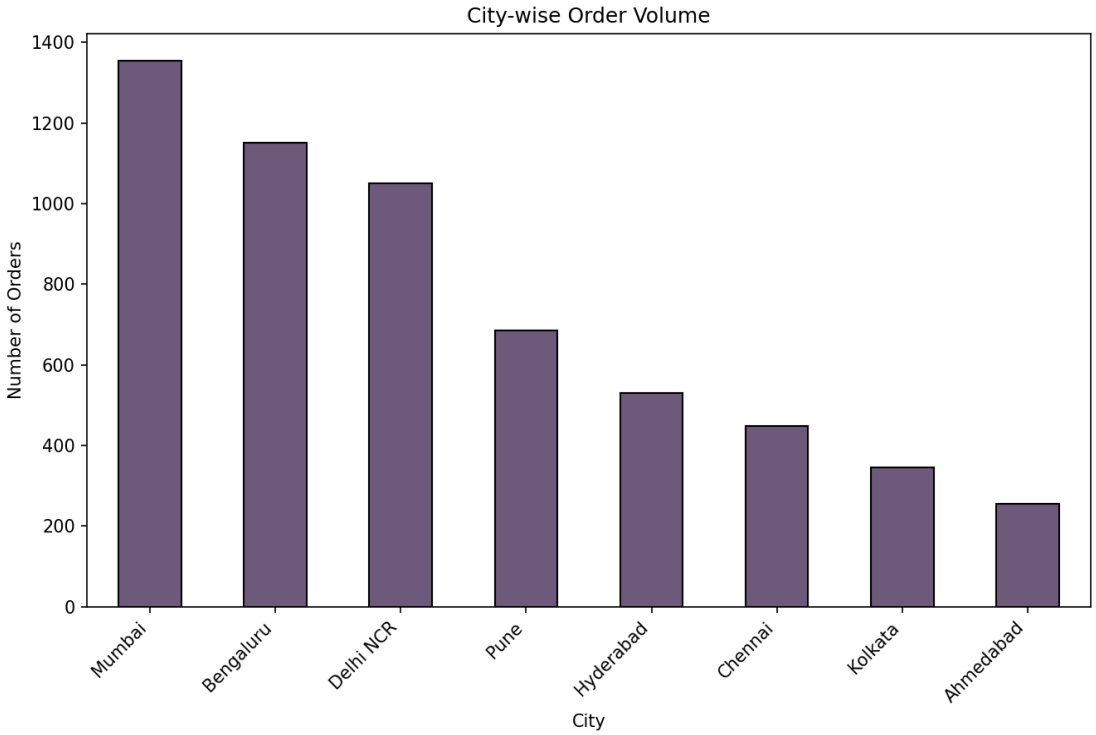


*Figure **5**A: Order volume by city*

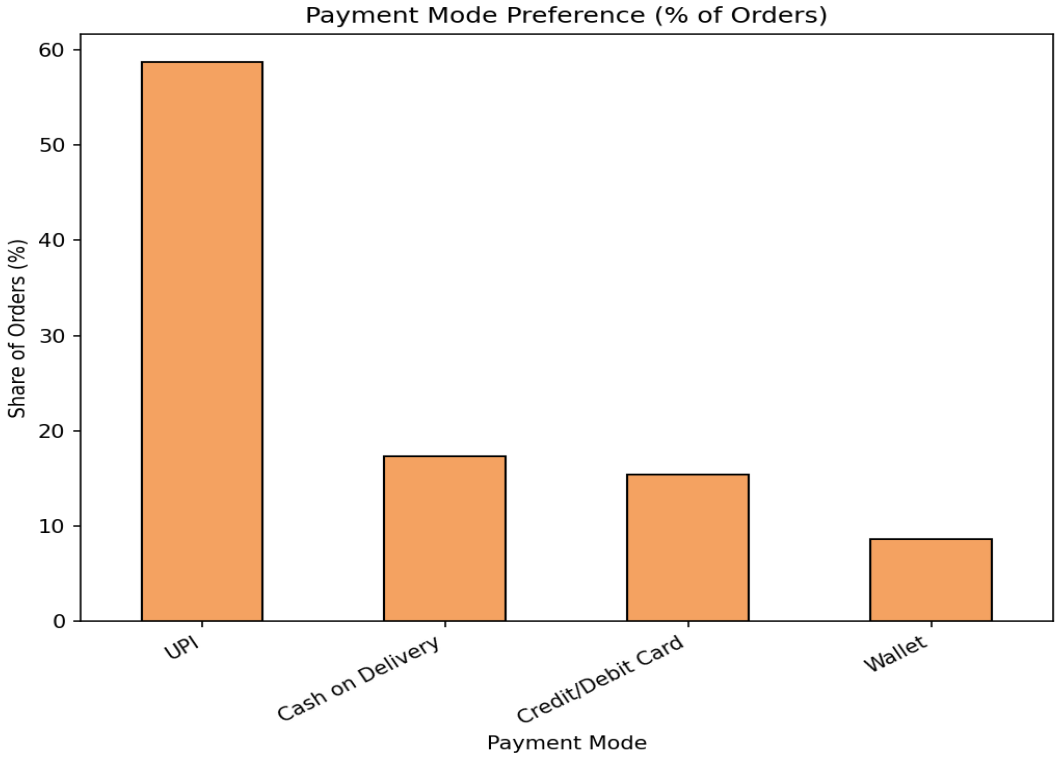


*Figure **5**B: Payment mode preference (% of orders)*

Figure **5**A shows the order volume recorded across the eight cities in the dataset, and Figure **5**B shows the share of orders by payment mode. The X-axis in Figure **5**A represents the city, and the Y-axis represents the total number of orders generated by customers in that city. In Figure **5**B, the X-axis represents the payment mode, and the Y-axis represents the percentage share of total orders.

Observation

The graph indicates that Mumbai generated the highest number of orders (**1,354** orders, **23.28%** of the total), making it Zepto's largest city segment in this dataset. Bengaluru ranked second (**1,150** orders, **19.77%**), followed by Delhi NCR (**1,050** orders, **18.05%**). Pune (**11.78%**), Hyderabad (**9.11%**), and Chennai (**7.70%**) contributed moderate shares, while Kolkata (**5.93%**) and Ahmedabad (**4.37%**) recorded the smallest shares of total order volume. On the payment side, UPI was by far the most preferred mode, accounting for **58.75%** of all orders, followed by Cash on Delivery (**17.26%**), Credit/Debit Card (**15.35%**), and Wallet (**8.63%**). The customer base also skews young: the **25**–**34** age group made up **37.62%** of orders and **18**–**24** made up **34.70%**, together accounting for nearly three-quarters of all orders.

Business Insight

The results suggest that Zepto should continue concentrating dark-store density and marketing spend in Mumbai, Bengaluru, and Delhi NCR, which together account for over **60%** of order volume, while using targeted city-specific promotions to grow share in lower-penetration cities such as Kolkata and Ahmedabad. The dominance of UPI (**58.75%**) confirms that UPI should remain the default, most

Summary of Key Metrics

Objective	Key Metric
Most ordered product	Amul Toned Milk **500**ml (**1,208** units sold)
Daily order volume trend	Average daily orders: ~**194**/day (range **145**–**325**)
Peak order hours	Peak order hour: **9** PM IST (evening); secondary peak **8**–**9** AM
Category ratings	Highest rated category: Fruits & Vegetables (**4.47**)
On-time delivery performance	Highest on-time category: Snacks & Munchies (**64.97%**)
City-wise engagement	Largest city: Mumbai (**23.28%** of orders)
Payment mode preference	Most preferred payment mode: UPI (**58.75%**)
Overall order value	Total revenue: **₹6,93,204.65** · Avg. order value: **₹119.19**
Order reliability	Delivered: **96.93%** · Cancelled: **3.07%**

How This Would Work at Real Scale

The dataset used in this report is synthetic and created for academic purposes, but the same descriptive analytics approach can be applied to Zepto's real order-management platform. In a production environment, millions of customers across hundreds of dark stores generate order data every minute through app browsing, cart additions, checkout, delivery tracking, and post-delivery ratings. Each interaction — adding an item to cart, selecting a payment method, receiving a delivery, or rating an order — would be recorded and analysed in near real time.

At real scale, this data would be collected across cities, dark stores, and delivery partners, and processed through cloud-based analytics and event-streaming systems. The platform could automatically track the most ordered products, daily and hourly demand trends, category-level delivery performance, and city-wise engagement. These insights would help Zepto optimise inventory placement at individual dark stores, delivery-partner scheduling, and city-specific marketing campaigns.

For example, if a specific dark store consistently sees a **9** PM demand spike for dairy and snacks, the platform could pre-position that inventory earlier in the evening and schedule additional riders for that slot. Similarly, category-level on-time delivery percentages — such as the low figure observed here for Meat & Seafood — would flag operational bottlenecks that need dedicated fulfilment workflows before they affect customer retention at scale.

Overall, descriptive analytics at real scale would provide real-time visibility into ordering behaviour, delivery performance, and regional demand, enabling Zepto to make faster, more effective, and more localised operational decisions.

Next Steps to Move From Descriptive → Predictive

The descriptive analytics performed in this report provides a strong foundation for future predictive analytics. The next step would be to use historical order data to forecast future customer behaviour, product demand, and delivery performance. Machine learning models could be used to predict which products are likely to run out of stock, which customers may churn, and which customers are most likely to upgrade to a paid membership or subscription plan. Predictive models could also estimate peak-hour order demand at the individual dark-store level, improve product recommendations, optimise delivery-partner allocation, and support new dark-store location planning.

The predictive analytics stage can focus on the following key areas:

- Predict future product demand using historical order trends, to guide inventory replenishment.
- Forecast peak order hours at the dark-store level for better rider scheduling and capacity planning.
- Identify customers at risk of churn to improve retention through targeted offers.
- Personalise product recommendations based on order history and category preferences.
- Predict category-level delivery-time risk (e.g., Meat & Seafood) to trigger proactive operational interventions.

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Dataset Citation

Synthetic Zepto Sales Dataset (**30** Days), generated for academic descriptive analytics purposes using simulated order behaviour, Indian city-wise customer preferences, and realistic quick-commerce engagement patterns.

Appendix

Appendix Item	Description
Appendix A	Sample of the synthetic Zepto order dataset (**30** days)
Appendix B	Python code used for data generation
Appendix C	Python code used for KPI calculations
Appendix D	Python code used for data visualization (Matplotlib)
Appendix E	Category-wise average customer rating table
Appendix F	Category-wise on-time delivery percentage table
Appendix G	City-wise order count table
Appendix H	Hour-wise order count table
Appendix I	Daily order volume trend data
Appendix J	Payment mode and age-group distribution tables

THANK

YOU